POLITECNICO
MILANO 1863

Design of an RC flat slab

CALCULATION REPORT

In partial fullfillment for the requirements of the Course of Advanced Strucural Design (MSc in Civil Engineering)

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1. Project Specifications

Building Description:

- Dimensions: Approximately 22 m x 67 m. (12 bays @ 5.5m and 5 bays @ 4.3m)
- Structure: 7 stories above ground with a one-story basement. (Inspired by ACI Design Handbook).

Structural Details:

- Horizontal structure: Reinforced Concrete (RC) flat slab supported by RC columns.
- Columns: Square cross-section (60 cm x 60 cm).
- Slab thickness: 18 cm.

2. Material Properties

- Concrete: Class C35/45 (fck = 35 MPa), Properties are according to (Eurocode2 § Table 3.1).
- Reinforcement Steel: Class B450C (fyk = 450 MPa) Properties are according to (Eurocode 2 § Annex C).

3. Assignment Scope

- Design reinforcement for a typical slab story and providing the drawings
- Provide top and bottom layers of reinforcement in both directions.
- Consider only gravity loads, excluding seismic or accidental loads.
- Addressing failure mechanisms such as conical collapse and punching shear using appropriate criteria.
- Verifying stress limits in reinforcement.
- Estimating deflections at the initial stage (t0) under SLS loads.

4. Load Cases

- ULS and SLS are according to the Table 1 of the Assignment file

Load Cases		
g1	4.5	kN / m^2
g2	0.8	kN / m^2
Q	4.8	kN / m^2
Load Combinations		
ULS	14.3	kN / m^2
SLS (characteristic)	10.1	kN / m^2

Figure 1

5. Structural Analysis

Internal forces (i.e. bending moments and shear) are derived from elastic analysis considering closed—form solution given by **Timoshenko**, and for the calculation MATLAB script such as the one given in Appendix A of the assignment is used. For the sake of validation, the internal forces derived from the mentioned solution should be compared to **Equivalent Frame Method (EFM)** also the reinforcement Design.

Timoshenko

Basically, the MATLAB code calculate deflection (w) at every point and then M_x, M_y, M_{xy} are calculated and after that Q_x, Q_y are derived.

a=4.30 [m] along x-direction and b=5.50 [m] along y-direction. The coordinate axis is located with its origine at the center of the plate

The deflection of the plate takes the form

$$w = \frac{qb^4}{384D}\left(1 - \frac{4y^2}{b^2}\right)^2 + A_0 + \frac{qa^3b}{2\pi^3 D}\sum_{m=2,4,6,\ldots}^{\infty} \frac{(-1)^{m/2}\cos\frac{m\pi x}{a}}{m^3 \sinh\alpha_m \tanh\alpha_m}$$
$$\left[\tanh\alpha_m \frac{m\pi y}{a}\sinh\frac{m\pi y}{a} - (\alpha_m + \tanh\alpha_m)\cosh\frac{m\pi y}{a}\right] \quad (l)$$

The constant A_0 can now be determined from the condition that the deflection vanishes at the corners of the plate. Hence

$$(w)_{x=a/2,y=b/2} = 0$$

and

$$A_0 = -\frac{qa^3b}{2\pi^3 D}\sum_{m=2,4,6,\ldots}^{\infty}\frac{1}{m^3}\left(\alpha_m - \frac{\alpha_m + \tanh\alpha_m}{\tanh^2\alpha_m}\right) \quad (m)$$

Figure 2

$$M_x = D(\chi_x + \nu\chi_y)$$

$$M_y = D(\chi_y + \nu\chi_x)$$

$$M_{xy} = D\frac{1-\nu}{2}\chi_{xy}$$

$$Q_x = \frac{\partial M_x}{\partial x} + \frac{\partial M_{yx}}{\partial y}$$

$$Q_y = \frac{\partial M_{xy}}{\partial x} + \frac{\partial M_y}{\partial y}$$

Figure 3

Concentrated reactions are acting at points of the support, and the moments calculated from Figure3 expressions become infinitely large. According to the provided solution by Timoshenko moments at corners can be calculated by the expressions below in which:

$$c = 0.57u\ (u\text{: square column dimension})\ , q = e^{\left(-\frac{\pi b}{a}\right)}$$

$$(M_x)_{x=a/2,y=b/2} = M_0 - \frac{qb^2}{4}\left[\frac{\nu}{3} + (1-\nu)\sum_{n=1}^{\infty}\frac{1}{\sinh^2\frac{m\pi b}{a}}\right]$$
$$(M_y)_{x=a/2,y=b/2} = M_0 + \frac{qb^2}{4}\left[-\frac{1}{3} + (1-\nu)\sum_{n=1}^{\infty}\frac{1}{\sinh^2\frac{n\pi b}{a}} + (1-\nu)\frac{a}{\pi b}\right] \quad (o)$$

In these expressions

$$M_0 = -\frac{qab}{4\pi}\left[(1+\nu)\log\frac{a}{2\pi c(1-q^2)^2(1-q^4)^2\cdots} + 1\right]$$

Figure 4

For the requested benchmark: the code was able to reproduce $w(0,0) = 2.993mm$ the calculated deflection according to the Table 57 by Timoshenko was $w(0,0) = 3.01m$ which results to 0.5% error

The results from the calculation are presented in the following Figures (5,6,7,8,9)

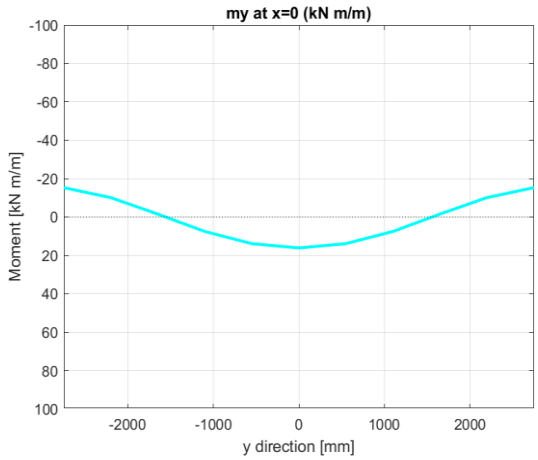


Figure 6

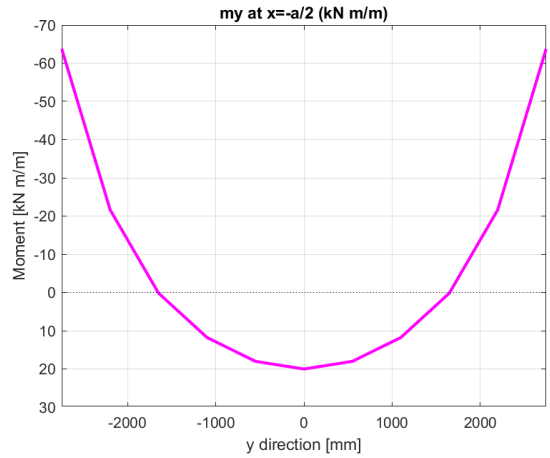


Figure 5

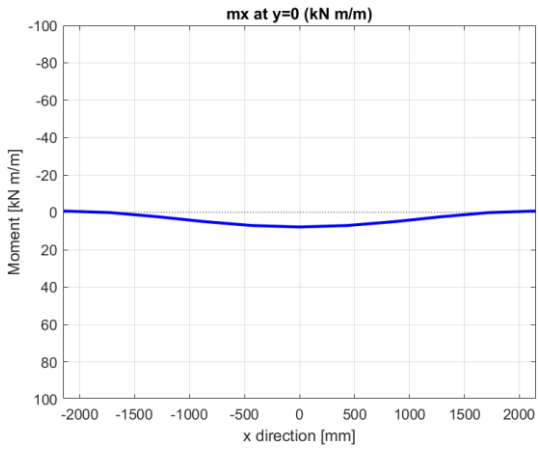


Figure 7

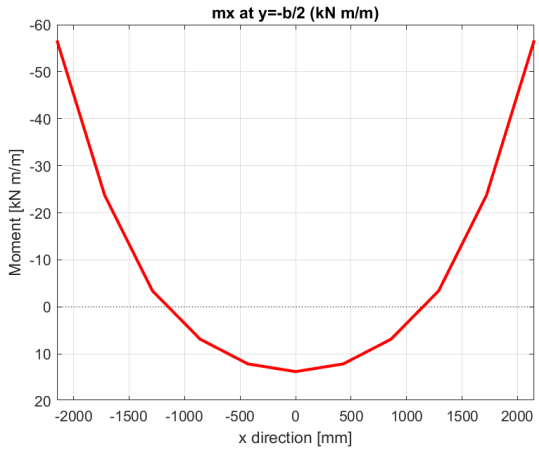


Figure 8

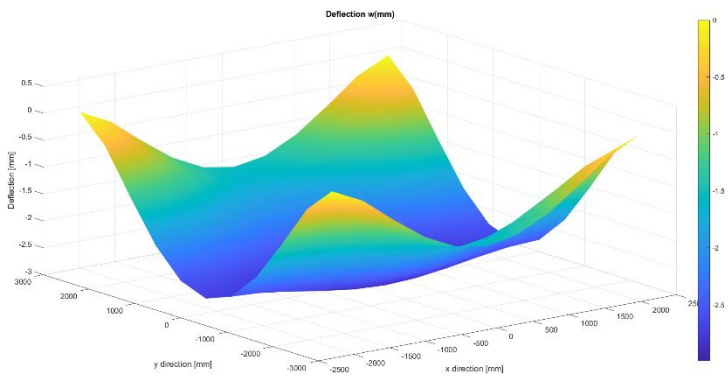


Figure 9

EFM

For the sake of validation Equivalent Frame method (EFM) was used. And for the calculation Ftool Program was adopted. For every direction 2 types of sub frames considered, an internal frame and an external frame. The frames were modeled like the Figures (10,11,12,13) below and the result are calculated as shown in the Figures

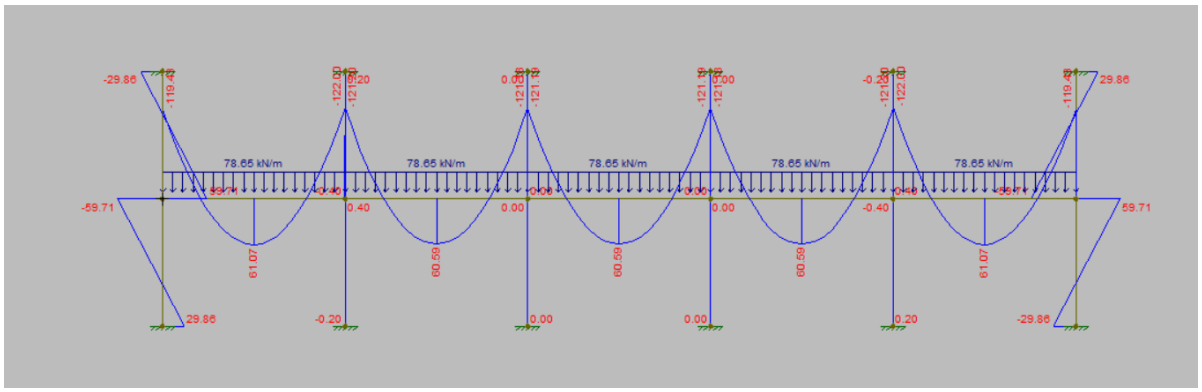


Figure 10-internal frame x direction

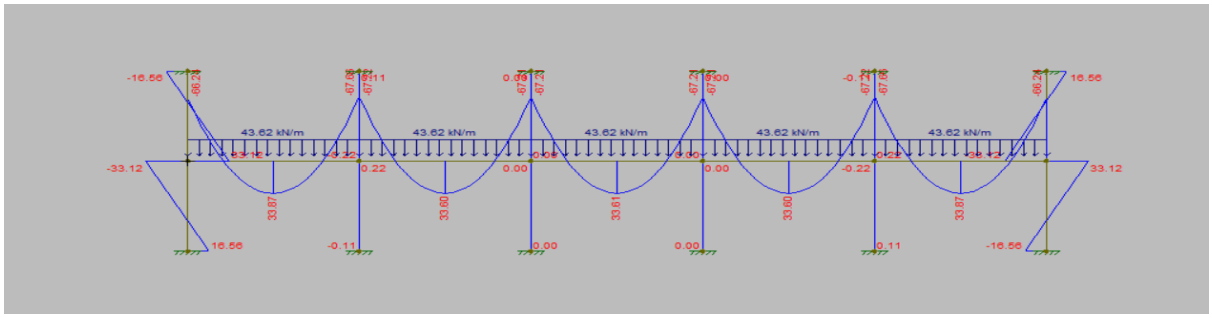


Figure 11-external frame x direction

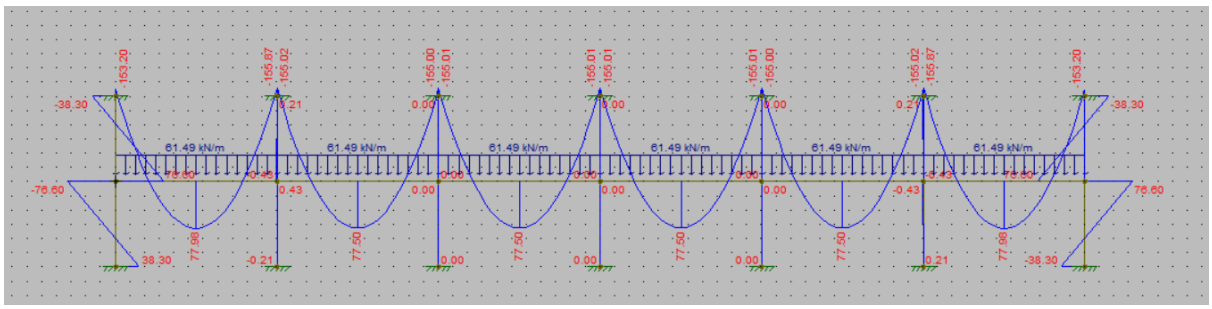


Figure 12- -internal frame y direction

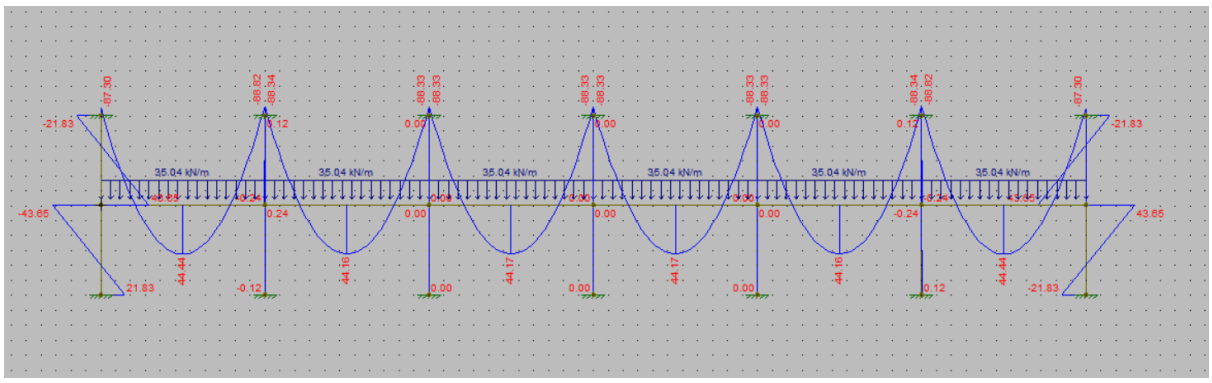


Figure 13- external frame y direction

Comparison

The results obtained from the script are **pointwise values**, meaning they give results at specific points rather than over a region. To compare the closed-form solution results with **EFM outcomes**, the pointwise values need to be **integrated over the relative length**. Three reference points were adopted for sake of comparison, Point A and B at the column's edge and point C at mid span (Figure 14)

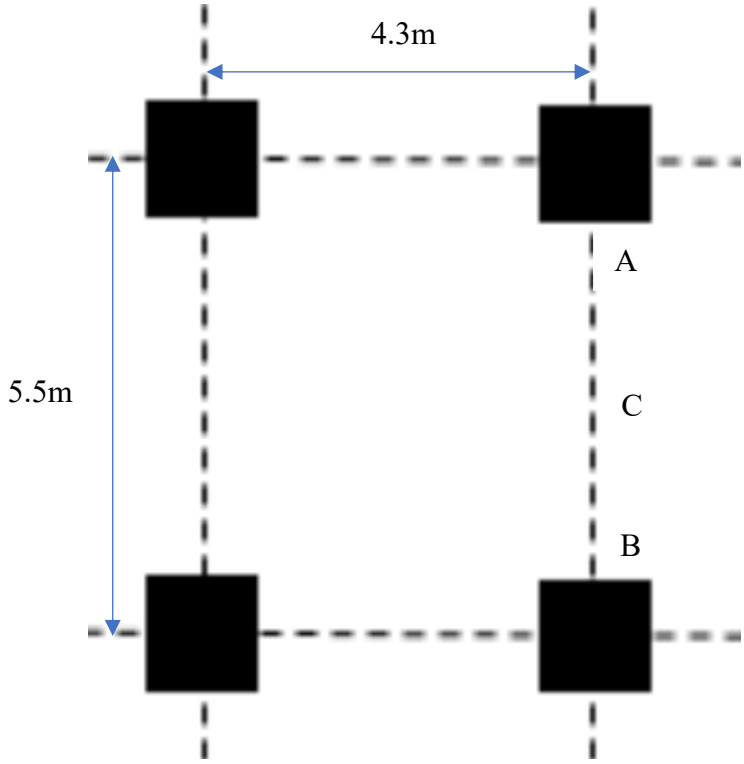


Figure 14

$Direction$	$Points$	$Timoshenko$	EFM	$Moment\ Sign$
		$kN.m$	$kN.m$	
x	A, B	99.94	121.19	$(-)$
	C	59.75	60.59	$(+)$
y	A, B	140.17	155.01	$(-)$
	C	77.50	77.50	$(+)$

Table 1

6. Reinforcement Design

Minimum and Maximum Reinforcement

According to EC2-4.4.1.2 the recommended structural class is S4. XC4 class was considered for the exposure condition of the building which results to the minimum cover of 30mm. reinforcement bar size is considered Ø12 and $d_x = 144\ mm\ , d_y = 132mm$

Minimum and maximum reinforcement is derived base on the EC-9.2.1.1

$$A_{s\,min} = 0.26\frac{f_{ctm}}{f_{yk}}b_t d\ not\ less\ than\ 0.0013b_t d\ \ , A_{s\,max} = 0.04A_c$$

Also, for considering the cracking moment minimum area of longitudinal tension reinforcement is calculated:

$$M_{cr} = \frac{f_{ctd}bh^2}{6}\ , A_{smin,cr} = \frac{M_{cr}}{0.9f_{yd}d}$$

$reinforcement\ for\ cracking\ moment$			
M_{cr}	$11.6\ kN.\frac{m}{m}$	M_{cr}	$11.6\ kN.\frac{m}{m}$
As_{cr} (1)	$227.9\frac{mm^2}{mm}$	As_{cr} (1)	$248.6\frac{mm^2}{mm}$

Table 2

$Minimum\ and\ maximum\ reinforcements\ \left(\frac{mm^2}{mm}\right)$			
$X\ direction$		$Y\ direction$	
A_{smin} (2)	178.0	A_{smin} (2)	163.2
A_{smin2} (3)	187.2	A_{smin2} (3)	171.6
$min(1,2,3)$	227.9	$min(1,2,3)$	248.6
ρ_s_min	0.13%	ρ_s_min	0.014%
A_{smax}	7200.0	A_{smax}	7200.0

Table 3

EC2-9.3.1.1 introduced the values of maximum spacing for principal reinforcement:

$$S_{max} = \min\{3h, 400mm\}\ and\ \ S_{max} = \min\{2h, 250mm\}$$

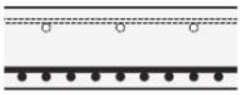
For the case of study, the maximum spacing is evaluated 250mm

Wood Armer Method

Reinforcement area (top/bottom) in both directions derived from the application of "Normal Moment Criterion" according to Wood and Armer. if we have the compression is not necessary to put reinforcement. But when we take into account the behavior at infinite time of the concrete, during the life time of the plate if we don't put steel in compression, we have a change of N.A. position and a decreasing of z_x or z_y. Also, this increases the deformability of our plate. so the minimum reinforcement is used when the reinforcement is not necessary. Because of symmetry only one quarter of the slab is designed ($x = 0\ to\ 2.15\ , y = 0\ to\ 2.75$)

BOTTOM REINFORCEMENT

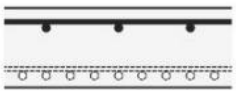
$$\begin{cases} m_{xu}^{+} = M_{xu} + \left|M_{xyu}\right| = \mathcal{M}_{xu}^{+} \\ m_{yu}^{+} = M_{yu} + \left|M_{xyu}\right| = \mathcal{M}_{yu}^{+} \end{cases} \quad \text{if } \mathcal{M}_{xu}^{+} \geq 0 \text{ and } \mathcal{M}_{yu}^{+} \geq 0$$



$$\begin{cases} m_{xu}^{+} = M_{xu} + \dfrac{M_{xyu}^{2}}{\left|M_{yu}\right|} \geq 0 \\ m_{yu}^{+} = 0 \end{cases} \quad \text{if } \mathcal{M}_{yu}^{+} < 0 \qquad \begin{cases} m_{xu}^{+} = 0 \\ m_{yu}^{+} = M_{yu} + \dfrac{M_{xyu}^{2}}{\left|M_{xu}\right|} \geq 0 \end{cases} \quad \text{if } \mathcal{M}_{xu}^{+} < 0$$

TOP REINFORCEMENT

$$\begin{cases} m_{xu}^{-} = M_{xu} - \left|M_{xyu}\right| = \mathcal{M}_{xu}^{-} \\ m_{yu}^{-} = M_{yu} - \left|M_{xyu}\right| = \mathcal{M}_{yu}^{-} \end{cases} \quad \text{if } \mathcal{M}_{xu}^{-} \leq 0 \text{ and } \mathcal{M}_{yu}^{-} \leq 0$$



$$\begin{cases} m_{xu}^{-} = M_{xu} - \dfrac{M_{xyu}^{2}}{\left|M_{yu}\right|} \leq 0 \\ m_{yu}^{-} = 0 \end{cases} \quad \text{if } \mathcal{M}_{yu}^{-} > 0 \qquad \begin{cases} m_{xu}^{-} = 0 \\ m_{yu}^{-} = M_{yu} - \dfrac{M_{xyu}^{2}}{\left|M_{xu}\right|} \leq 0 \end{cases} \quad \text{if } \mathcal{M}_{xu}^{-} > 0$$

Figure 15

$m_x\ (kN\frac{m}{m})$						$m_y\ \left(kN\frac{m}{m}\right)$					
7.82	7.04	4.98	2.34	0.12	−0.74	16.12	16.46	17.38	18.59	19.62	20.02
8.21	7.31	4.89	1.69	−1.06	−2.16	13.89	14.22	15.15	16.42	17.56	18.02
9.48	8.26	4.79	−0.15	−4.84	−6.84	7.55	7.78	8.52	9.76	11.14	11.78
11.68	10.10	5.29	−2.61	−11.82	−16.48	−1.50	−1.70	−2.08	−1.98	−0.79	0.19
14.15	12.45	7.00	−3.49	−21.32	−36.68	−10.10	−11.16	−14.29	−18.99	−22.60	−21.62
13.77	12.14	6.88	−3.41	−23.72	−56.66	−15.29	−16.88	−22.03	−32.18	−52.37	−63.74

Table 4

$m_{xy}\ \left(kN\frac{m}{m}\right)$					
0.00	0.00	0.00	0.00	0.00	0.00
0.00	−1.05	−1.79	−1.91	−1.25	0.00
0.00	−2.12	−3.76	−4.24	−2.92	0.00
0.00	−2.86	−5.47	−7.00	−5.57	0.00
0.00	−2.31	−4.91	−7.89	−9.51	0.00
0.00	0.00	0.00	0.00	0.00	0.00

Table 5

$\mathcal{M}_{xu}^{+}, \mathcal{M}_{xu}^{-}\left[kN\frac{m}{m}\right]$						$\mathcal{M}_{yu}^{+}, \mathcal{M}_{yu}^{-}\left[kN\frac{m}{m}\right]$					
7.82	7.04	4.98	2.34	0.12	0.74	16.12	16.46	17.38	18.59	19.62	20.02
8.21	6.27	3.10	0.22	2.31	2.16	13.89	13.17	13.37	14.51	16.31	18.02
9.48	6.14	1.03	4.39	7.76	6.84	7.55	5.66	4.76	5.52	8.22	11.78
11.68	7.23	0.18	9.61	17.39	16.48	1.50	4.57	7.55	8.97	6.37	0.19
14.15	10.13	2.10	11.38	30.83	36.68	10.10	13.47	19.19	26.88	32.11	21.62
13.77	12.14	6.88	3.41	23.72	56.66	15.29	16.88	22.03	32.18	52.37	63.74

Table 6

Calculated reifnorcements											
$a_{sx}\ \left[\frac{mm^2}{m}\right]$						$a_{sy}\ \left[\frac{mm^2}{m}\right]$					
154.12	138.90	98.18	46.10	2.46	14.58	317.79	324.54	342.80	366.57	386.83	394.83
161.88	164.91	131.58	71.08	45.51	42.52	273.83	301.12	334.05	361.48	370.81	355.30
186.99	204.72	168.53	86.64	153.06	134.89	148.85	195.29	242.05	276.06	277.17	232.30
230.31	255.58	212.12	189.51	342.89	324.92	29.63	90.07	148.85	176.96	125.53	3.79
278.99	291.13	234.78	224.37	607.86	723.21	199.07	265.61	378.41	530.10	633.17	426.32
271.61	239.32	135.65	67.34	467.66	1117.23	301.46	332.91	434.37	634.63	1032.75	1256.92

Table 7

Lower reinforcement Design											
$\rho_{sx\ design}$ [%] (+)						$\rho_{sy\ design}$ [%] (+)					
0.13%	0.13%	0.13%	0.13%	0.13%	0.13%	0.18%	0.18%	0.19%	0.20%	0.21%	0.22%
0.13%	0.13%	0.13%	0.13%	0.13%	0.13%	0.15%	0.17%	0.19%	0.20%	0.21%	0.20%
0.13%	0.13%	0.13%	0.13%	0.13%	0.13%	0.14%	0.14%	0.14%	0.15%	0.15%	0.14%
0.13%	0.14%	0.13%	0.13%	0.13%	0.13%	0.14%	0.14%	0.14%	0.14%	0.14%	0.14%
0.15%	0.16%	0.13%	0.13%	0.13%	0.13%	0.14%	0.14%	0.14%	0.14%	0.14%	0.14%
0.15%	0.13%	0.13%	0.13%	0.13%	0.13%	0.14%	0.14%	0.14%	0.14%	0.14%	0.14%

Table 8

Upper reinforcement Design											
$\rho_{sx\ design}$ [%] (−)						$\rho_{sy\ design}$ [%] (−)					
0.13%	0.13%	0.13%	0.13%	0.13%	0.13%	0.14%	0.14%	0.14%	0.14%	0.14%	0.14%
0.13%	0.13%	0.13%	0.13%	0.13%	0.13%	0.14%	0.14%	0.14%	0.14%	0.14%	0.14%
0.13%	0.13%	0.13%	0.13%	0.13%	0.13%	0.14%	0.14%	0.14%	0.14%	0.14%	0.14%
0.13%	0.13%	0.13%	0.13%	0.19%	0.18%	0.14%	0.14%	0.14%	0.14%	0.14%	0.14%
0.13%	0.13%	0.13%	0.13%	0.34%	0.40%	0.14%	0.15%	0.21%	0.29%	0.35%	0.24%
0.13%	0.13%	0.13%	0.13%	0.26%	0.62%	0.17%	0.18%	0.24%	0.35%	0.57%	0.70%

Table 9

EFM With Redistirbution

From the Ftool model the moments derived for an internal panel were:

$M_x^- = 121.19\ kN.m$	$M_y^- = 155.01\ kN.m$
$M_x^+ = 60.59\ kN.m$	$M_y^+ = 77.50\ kN.m$

Table 10

According to the EC 5.5

In continuous beams or slabs which:

a) are predominantly subject to flexure

b) have the ratio of the lengths of adjacent spans in the range of 0,5 to 2, redistribution of bending moments may be carried out without explicit check on the rotation capacity, provided that:

$\delta \geq k_1 + k_2 x_u/d$ for $f_{ck} \leq 50$ MPa (5.10a)

$\delta \geq k_3 + k_4 x_u/d$ for $f_{ck} > 50$ MPa (5.10b)

$\geq k_5$ where Class B and Class C reinforcement is used (see Annex C)
$\geq k_6$ where Class A reinforcement is used (see Annex C)

Where:
δ is the ratio of the redistributed moment to the elastic bending moment
x_u is the depth of the neutral axis at the ultimate limit state after redistribution

d is the effective depth of the section

Note: The values of k_1, k_2, k_3, k_4, k_5 and k_6 for use in a Country may be found in its National Annex. The recommended value for k_1 is 0,44, for k_2 is 1,25(0,6+0,0014/ε_{cu2}), for k_3 = 0,54, for k_4 = 1,25(0,6+0,0014/ε_{cu2}), for k_5 = 0,7 and k_6 = 0,8. ε_{cu2} is the ultimate strain according to Table 3.1.

Figure 16

Because of the bars having Class C: $\delta \geq 0.7$ →Redistribution factor of $\delta = 0.8$ is considered

$M_x^- = 96.95\ kN.m$	$M_y^- = 124\ kN.m$
$M_x^+ = 84.83\ kN.m$	$M_y^+ = 108.5\ kN.m$

Table 11

The pre-dimensioning amount of reinforcement is based on this formula:

$$a_s = \frac{M_{red}}{0.9 d f_{yd}} \left(\frac{mm^2}{mm}\right)$$

$Pre-dimensioned\ reinforcement\ \left(\frac{mm^2}{mm}\right)$			
a_{sy}^-	485.00	a_{sx}^-	444.59
a_{sy}^+	424.38	a_{sx}^+	389.00

Table 12

$Designed\ reinforcement$			
a_{sx}^-	1Ø12/25	ρ_{sx}^- [%]	0.25%
a_{sx}^+	1Ø12/25	ρ_{sx} [%]	0.25%
a_{sy}^-	1Ø12/20	ρ_{sy}^- [%]	0.31%
a_{sy}^+	1Ø12/20	ρ_{sy}^+ [%]	0.31%

Table 13

Reinforcement Comparison

Cross Section	***Wood − Armer***	***EFM + Redistribution***	***Moment Sign***
Panel Boundary (along x direction)	0.13%	0.25%	(+)
	0.23%	0.25%	(−)
Panel center (along x direction)	0.13%	0.25%	(+)
	0.13%	0.25%	(−)
Panel center (along y direction)	0.15%	0.31%	(+)
	0.14%	0.31%	(−)
Panel boundary (along y direction)	0.16%	0.31%	(+)
	0.25%	0.31%	(−)

Table 14

7. Conical Collapse Mechanism

Considering conical collapse mechanisms for slab supported by regular grid of columns, comparison between analytically—derived ultimate loads (upper bound) and nominal ULS load should is for each type of column

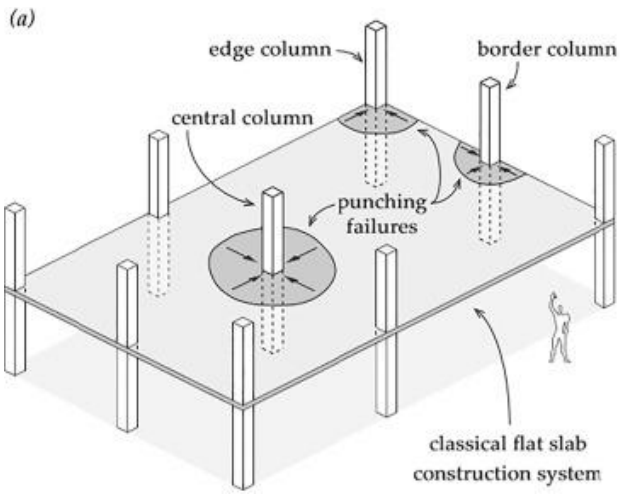


Figure 17

Due to identical properties 4 type of columns are considered

1-interior column 2-edge column in x direction 3- edge column along y direction 4 -corner column

$$P_{u,}conical = \alpha(m_{xu} + m_{yu})$$

m_{xu} , m_{yu} are calculated based on fully plastic rigid model

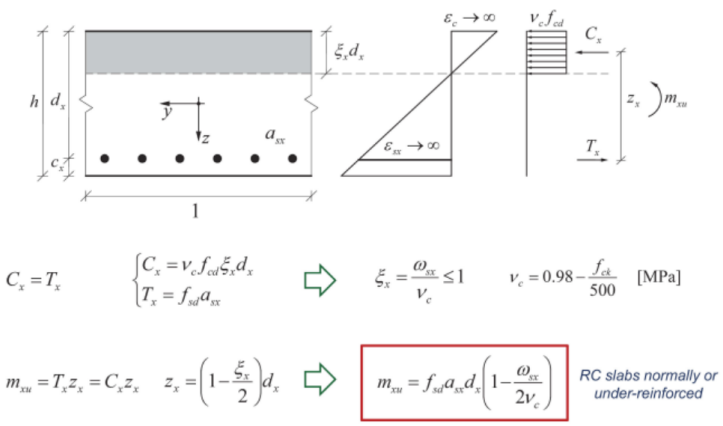


Figure 18

Furthermore, an estimate of the rotation (ψ) is given according to the following expression adapted from Muttoni

$$\psi = \frac{r_s}{d} \frac{f_{yd}}{E_s} \left(\frac{P_{uls}}{P_{u,conical}} \right)^{1.5}$$

Interior column			Edge column along x direction		
α	2π	$-$	α	π	$-$
m_{xu}	25.47	$\left(kN\frac{m}{m}\right)$	m_{xu}	25.47	$\left(kN\frac{m}{m}\right)$
m_{yu}	29.18	$\left(kN\frac{m}{m}\right)$	m_{yu}	29.18	$\left(kN\frac{m}{m}\right)$
$P_{u,conical}$	343.39	(kN)	$P_{u,conical}$	171.70	(kN)
A	23.65	(m^2)	A	11.83	(m^2)
$q_{u,conicnal}$	14.52	$\left(\frac{kN}{m^2}\right)$	$q_{u,conicnal}$	14.52	$\left(\frac{kN}{m^2}\right)$
q_{ULS}	14.30	$\left(\frac{kN}{m^2}\right)$	q_{ULS}	14.30	$\left(\frac{kN}{m^2}\right)$
P_{ULS}	338.20	(kN)	P_{ULS}	169.10	(kN)
Ψ	0.016	$-$	Ψ	0.016	$-$

Table 15

Corner Column			Edge column along y direction		
α	$\frac{\pi}{2}$	$-$	α	π	$-$
m_{xu}	25.47	$\left(kN\frac{m}{m}\right)$	m_{xu}	25.47	$\left(kN\frac{m}{m}\right)$
m_{yu}	29.18	$\left(kN\frac{m}{m}\right)$	m_{yu}	29.18	$\left(kN\frac{m}{m}\right)$
$P_{u,conical}$	85.85	(kN)	$P_{u,conical}$	171.70	(kN)
A	5.91	(m^2)	A	11.83	(m^2)
$q_{u,conicnal}$	14.52	$\left(\frac{kN}{m^2}\right)$	$q_{u,conicnal}$	14.52	$\left(\frac{kN}{m^2}\right)$
q_{ULS}	14.30	$\left(\frac{kN}{m^2}\right)$	q_{ULS}	14.30	$\left(\frac{kN}{m^2}\right)$
P_{ULS}	84.55	(kN)	P_{ULS}	169.10	(kN)
Ψ	0.016	$-$	Ψ	0.016	$-$

Table 16

8. Punching Shear

The punching shear resistance, at ULS is checked according to Muthoni's criterion

For each point at a distance d/2 from column's perimeter, being d the average effective depth of the slab at column's face, average tangential stresses (τ_{sd}) should be lower than tangential design resistance (τ_{rd})

$$(\tau_{rd}) = \frac{\tau_{RK}}{\gamma_c} \quad ,\tau_{RK} = \frac{0.66\sqrt{f_c}}{1 + 20\psi.\frac{d}{32}} \qquad ,\gamma_c = 1.5 \quad \tau_{ed} = \frac{qA_{trib}}{u_i d}\beta$$

β *is* 1.15 *for interior columns,* 1.4 *for edge columns, and* 1.5 *for corner columns*

Interior column			Corner column		
A	23.65	$[m^2]$	A	5.91	$[m^2]$
q_{ULS}	14.30	$[\frac{kN}{m^2}]$	q_{ULS}	14.30	$[\frac{kN}{m^2}]$
V_{ed}	338.20	$[kN]$	V_{ed}	84.55	$[kN]$
u_i	2833.54	$[mm]$	u_i	1308.38	$[mm]$
d	138.00	$[mm]$	d	138.00	$[mm]$
β	1.15	$[-]$	β	1.5	$[-]$
f_{ck}	35.00	$[\frac{N}{mm^2}]$	f_{ck}	35.00	$[\frac{N}{mm^2}]$
ψ	0.016	$[-]$	ψ	0.016	$[-]$
d_{avg}	138.00	$[mm]$	d_{avg}	138.00	$[mm]$
τ_{rk}	1.64	$[\frac{N}{mm^2}]$	τ_{rk}	1.64	$[\frac{N}{mm^2}]$
τ_{rd}	1.09	$[\frac{N}{mm^2}]$	τ_{rd}	1.09	$[\frac{N}{mm^2}]$
τ_{ed}	0.99	$[\frac{N}{mm^2}]$	τ_{ed}	0.70	$[\frac{N}{mm^2}]$
$\frac{\tau_{ed}}{\tau_{rd}}$	0.91	$[-]$	$\frac{\tau_{ed}}{\tau_{rd}}$	0.64	$[-]$
Column along x direction			Column along y direction		
A	11.83	$[m^2]$	A	11.83	$[m^2]$
q_{ULS}	14.30	$[\frac{kN}{m^2}]$	q_{ULS}	14.30	$[\frac{kN}{m^2}]$
V_{ed}	169.10	$[kN]$	V_{ed}	169.10	$[kN]$
u_i	2016.77	$[mm]$	u_i	2016.77	$[mm]$
d	138.00	$[mm]$	d	138.00	$[mm]$
β	1.40	$[-]$	β	1.40	$[-]$
f_{ck}	35.00	$[\frac{N}{mm^2}]$	f_{ck}	35.00	$[\frac{N}{mm^2}]$
ψ	0.016	$[-]$	ψ	0.016	$[-]$
d_{avg}	138.00	$[mm]$	d_{avg}	138.00	$[mm]$
τ_{rk}	1.64		τ_{rk}	1.64	
τ_{rd}	1.09	$[\frac{N}{mm^2}]$	τ_{rd}	1.09	$[\frac{N}{mm^2}]$
τ_{ed}	0.85	$\left[\frac{N}{mm^2}\right]$	τ_{ed}	0.85	$\left[\frac{N}{mm^2}\right]$
$\frac{\tau_{ed}}{\tau_{rd}}$	0.78	$[-]$	$\frac{\tau_{ed}}{\tau_{rd}}$	0.78	$[-]$

Table 17

9. SLS

Stress check

Stresses in steel reinforcement at SLS is checked according to Eurocode 2 (2004)[3]§7.2 ($\sigma < 0.8 \cdot fyk \approx 360$ MPa). Stress demands is evaluated for SLS (characteristic) load combination.

X direction EFM					
Lower reinforcement			*Upper reinforcement*		
Mx	7.78	$[kN*m/m]$	Mx	15.56	$[kN*m/m]$
a_{sx}^{+}	452.00	$[mm^2/m]$	a_{sx}^{+}	452.00	$[mm^2/m]$
d_x^{+}	0.144	$[m]$	d_x^{+}	0.144	$[m]$
a_{sx}^{-}	452.00	$[mm2/m]$	a_{sx}^{-}	452.00	$[mm^2/m]$
d_x^{-}	0.036	$[m]$	d_x^{-}	0.036	$[m]$
x	26.61	$[mm]$	x	26.61	$[mm]$
Jid	44998600	$[mm^4]$	Jid	44998600	$[mm^4]$
σ_{sx}^{+}	125.38	$[MPa]$	σ_{sx}^{+}	250.76	$[MPa]$
σ_{lim}	360.00	$[MPa]$	σ_{lim}	360.00	$[MPa]$
X direction Temoshenko					
Lower reinforcement			*Upper reinforcement*		
Mx	7.67	$[kN*m/m]$	Mx	12.83	$[kN*m/m]$
a_{sx}^{+}	452.00	$[mm^2/m]$	a_{sx}^{+}	452.00	$[mm^2/m]$
d_x^{+}	0.144	$[m]$	d_x^{+}	0.144	$[m]$
a_{sx}^{-}	452.00	$[mm2/m]$	a_{sx}^{-}	452.00	$[mm^2/m]$
d_x^{-}	0.036	$[m]$	d_x^{-}	0.036	$[m]$
x	26.61	$[mm]$	x	26.61	$[mm]$
Jid	44998600	$[mm^4]$	Jid	44998600	$[mm^4]$
σ_{sx}^{+}	123.63	$[MPa]$	σ_{sx}^{+}	206.79	$[MPa]$
σ_{lim}	360.00	$[MPa]$	σ_{lim}	360.00	$[MPa]$

Table 18

Such calculations are done for y direction an complete results are presented

Direction	*Points*	*Timoshenko*	*EFM*	*Moment Sign*
		MPA	*MPA*	
x	A, B	206.79	250.76	$(-)$
	C	123.63	125.38	$(+)$
y	A, B	315.14	348.50	$(-)$
	C	174.25	174.25	$(+)$

Table 19

Deflection

An estimate of the maximum deflection, for an interior panel, at initial stage (t0)—e.g. for proof-loading purposes—is given considering the SLS (characteristic) load combination. Although long-term effects may be critical for the slab to be designed, they are out of the scope of the presented results.

$$\frac{b}{a} = \frac{5.5}{4.3} = 1.28 - interpolation\ from\ table \rightarrow \alpha = 0.003956\ , \beta = 0.018233 \rightarrow w = 2.12\ mm$$

$$it\ is\ worth\ mentioning\ that\ using\ MATLAB\ code\ deflection\ was\ drived\ 2.11mm$$

b/a	$w = \alpha \frac{qb^4}{D}$ α	$M_x = \beta q b^2$ β	$M_y = \beta_1 q b^2$ β_1
1	0.00581	0.0331	0.0331
1.1	0.00487	0.0261	0.0352
1.2	0.00428	0.0210	0.0363
1.3	0.00387	0.0175	0.0375
1.4	0.00358	0.0149	0.0384
1.5	0.00337	0.0131	0.0387
2.0	0.00292	0.0092	0.0411
∞	0.00260	0.0083	0.0417

Table 20

10. Lap Length and Anchorage Length

anchorage of longitudinal reinforcement is based on the EC2– §8.4

The calculation of the required anchorage length shall take into consideration the type of steel and bond properties of the bars. (2) The basic required anchorage length,l_b, rqd, for anchoring the force $A_s \times \sigma_{sd}$ in a straight bar assuming constant bond stress equal to f_{bd} follows from:

$$l_{b,rqd} = \left(\frac{\phi}{4}\right)\left(\frac{\sigma_{sd}}{f_{bd}}\right)$$

$$where\ f_{bd} = 2.25\ \eta_1 \eta_2 f_{ctd}$$

$$f_{ctd} = \frac{(\alpha_{ct} f_{ctk,0.05})}{\gamma_c}$$

Design anchorage length is calculated from the provided equation $l_{b,d}$

$$l_{b,d} = \alpha_1 \alpha_2 \alpha_3 \alpha_4 \alpha_5 l_{b_{req}} \geq l_{b,min}$$

$$where \begin{cases} in\ tension\ \ l_{b,min} \geq \max(0{,}3 l_{b,req}; 10\varphi; 100mm) \\ in\ compression\ \ l_{b,min} \geq \max\{0.6 l_b, req; 10\phi m 100mm\} \end{cases}$$

For the lap length EC uses equations below:

$$l_{b,d} = \alpha_1 \alpha_2 \alpha_3 \alpha_4 \alpha_5 \alpha_6 l_{b_{req}} \geq l_{b,min}$$

$$l_{0,min} = \max\{0.3\alpha_6\ l_{b,req}\ ; 15\phi, 200\ mm\}$$

$Coefficients$		$Anchorage\ length$	(mm)
η_1	1.00	$l_{b,rqd}$	355.73
η_2	1.00	$compression\ \ l_{b,min}$	120.00
α_{ct}	1.00	$tension\ \ l_{b,min}$	213.27
γ_c	1.50	l_{bd}	192.83
f_{ctm}	$2.20[MPa]$	$Lap\ length$	(mm)
α_1	1.00	l_0	275.48
c_d	$30.00[mm]$	l_{0min}	200.00
α_2	0.78	$l_{0,d}$	300.00
α_3	1.00		
α_4	0.70		
α_5	1.00		
α_6	1.00		
σ_{sd}	$391.30[MPa]$		
f_{ctd}	$1.47[MPa]$		
f_{bd}	$3.30[MPa]$		
Ø	$12.00[mm]$		

Table 21